

LESSON-1

ATOMIC STRUCTURE

DAY-1

Week-3

: First Excited State (Taken $n=2$)

: Second ($n=3$)

$$E = mc^2$$

$$1 \text{ Joule} = 10^7 \text{ erg}$$

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$$

$$1 \text{ pm} = 10^{-12} \text{ m}, 1 \text{ \AA} = 10^{-10} \text{ m}$$

$$1 \text{ nm} = 10^{-9} \text{ m}$$

Electrostatic force/Coulombic force

$$= K \frac{q_1 q_2}{r^2} = \frac{mv^2}{r}$$

$$PE = \frac{K q_1 q_2}{r}$$

frequency...

$$h : 6.62 \times 10^{-27} \text{ erg}$$

$$E = \frac{hc}{\lambda} = h\nu \quad (\because h \rightarrow \text{Planck's Const} : 6.6 \times 10^{-34} \text{ J})$$

$$\text{Quantum Yield} = \frac{\text{NO. of Molecules reacting}}{\text{NO. of Quanta Absorb.}}$$

$$mvr = n \frac{h}{2\pi} \quad \boxed{2\pi r = n\lambda}$$

$$R_N = R_\infty [A]^{1/3}$$

$$r_n = a_0 \times \frac{n^2}{Z}$$

Bohr : Single Electron Species (or) Hydrogen Like Species...
(is only Applicable)

$$\boxed{\frac{KZe^2}{r^2} = \frac{mv^2}{r}}$$

$$\text{Energies: } E = -\frac{2\pi^2 me^4 K^2 Z^2}{n^2 h^2} = -\frac{2\pi^2 me^4 Z^2}{16\pi^2 \epsilon_0^2 n^2 h^2} \propto \frac{Z^2}{n^2}$$

$$E = -3136 \times \frac{Z^2}{n^2} \text{ Kcal/mol}$$

$$E = -13.6 \times \frac{Z^2}{n^2} \text{ eV/atom.}$$

$$E = -1312 \times \frac{Z^2}{n^2} \text{ KJ/mole.}$$

$$\text{Total Energy} = -KE = \frac{PE}{2}$$

$$\text{Ionization Energy: } 13.6 \times Z^2$$

$$\Delta E = E_2 - E_1 = -13.6 Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] \quad E_n = \frac{-13.6 \text{ eV}}{n^2}$$

$$\text{Velocity: } v = \frac{2\pi e^2 k z}{nh} \propto \frac{Z}{n}$$

$$v = 2.18 \times 10^6 \frac{Z}{n} \text{ m/sec}$$

$$v = 2.18 \times 10^8 \frac{Z}{n} \text{ cm/sec}$$

Radius:

$$r = \frac{n^2 h^2}{4\pi^2 m k z} \propto \frac{n^2}{Z}$$

$$r = 0.529 \times \frac{n^2}{Z} \text{ \AA}, \quad r = 52.9 \times \frac{n^2}{Z} \text{ pm}$$

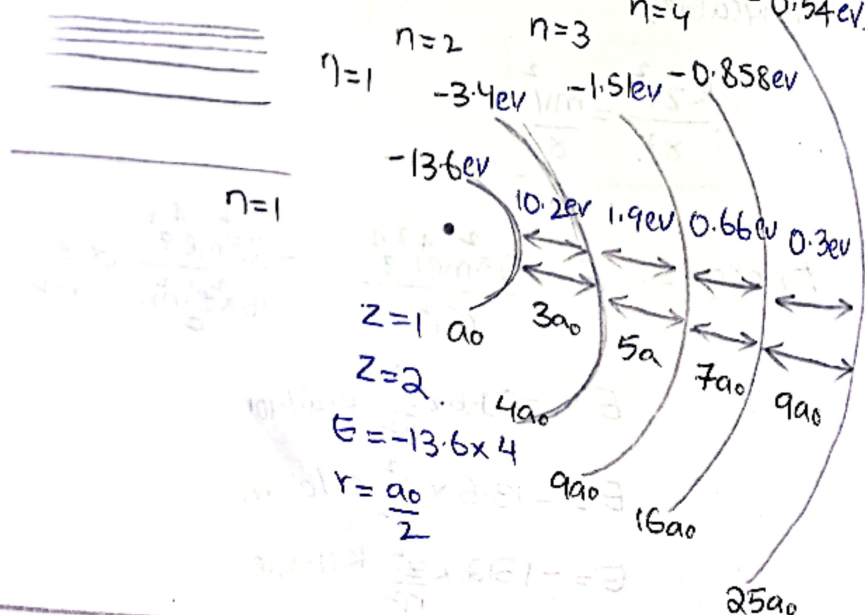
$$\text{Orbital Frequency: } \frac{v}{2\pi r} = 0.657 \times 10^{16} \frac{Z^2}{n^3}$$

$$f \propto \frac{Z^2}{n^3}$$

$$\text{Time per } \propto \frac{n^3}{Z^2}$$

$$= 1.52 \times 10^{-16} \frac{n^3}{Z^2}$$

Spectral Lines:



Laymen	$n_1=1$	UV	$n_2=2,3,4$
Balmer	$n_1=2$	Visible	$n_2=3,4,5$
Paschen	$n_1=3$	IR	$n_2=4,5,6$
Bracket	$n_1=4$	IR	$n_2=5,6,7$
Pfund	$n_1=5$	IR	$n_2=6,7,8$
Humphrey	$n_1=6$	IR	$n_2=7,8,9$

$$\text{no. of Spectral Lines} = \frac{n(n-1)}{2}$$

where

$n \rightarrow$ Orbit Value.

Ground State $\neq 1$

$$= \frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2} \quad \begin{matrix} n_2 = 6 \\ n_1 = 2 \end{matrix}$$

$$= \frac{(6-2)(6-2+1)}{2}$$

$$= \frac{(4)(5)}{2} = 10.$$

$$\begin{matrix} 6 \rightarrow 5, 4, 3, 2 \\ 5 \rightarrow 4, 3, 2 \\ 4 \rightarrow 3, 2 \\ 3 \rightarrow 2 \end{matrix} \quad (10)$$

wave number $\leftarrow$ $\frac{1}{\lambda} = \bar{\nu} = RZ^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$

wave length $\leftarrow$

$$R \rightarrow 109677 \text{ cm}^{-1} \text{ Nm/Coulomb} \rightarrow 1.097 \times 10^7 \text{ m}^{-1}$$

(Rydberg Constant)

H_α / First Line / Longest λ / Shortest ν / Low Energy

H_β / Second Line

$$\boxed{n_2 = n_1 + 1}$$

$$H_\beta = n_2 = n_1 + 2$$

$$H_\gamma = n_2 = n_1 + 3$$

Proton: q, m

Deuteron: $q, 2m$

α Particle $2q, 4m$.

Atomicity = Total No. of Atoms in a Molecule ...

Lyman:

$$\lambda_{\text{Max}} = 1216 \text{ \AA}$$

$$\lambda_{\text{Min}} = 912 \text{ \AA}$$

Balmer:

$$\lambda_{\text{Max}} = 6563 \text{ \AA}$$

$$\lambda_{\text{Min}} = 3648 \text{ \AA}$$

Photo Electric Effect:

$$E = E_0 + \frac{1}{2}mv^2$$

$$h\nu = h\nu_0 + \frac{1}{2}mv^2$$

$$KE = h[\nu - \nu_0]$$

$$\text{Stopping Potential} \Rightarrow V_0 = \frac{KE_{\text{Max}}}{e}$$

where,

$e \rightarrow$ Charge.

$$V \propto KE$$

no. of incident rays $\propto$ no. of P. $\bar{e} \propto P \cdot E \propto$ Intensity
Photo. (Photo Current Electricity)

De Broglie's Wavelength:

$$\lambda = \frac{h}{mv} = \frac{h}{p}$$

$$2\pi r_n = n\lambda$$

$$\lambda = \frac{h}{\sqrt{2m \cdot KE}} = \frac{h}{\sqrt{2mqV}}$$

$$\lambda_e = \frac{12.27}{\sqrt{V}} \times 10^{-10} \text{ m} \quad [\because V \rightarrow \text{Potential Difference}]$$

$$\lambda_{\text{Photon}} = \frac{0.286}{\sqrt{V}} \times 10^{-10} \text{ m}$$

$$\lambda_{\text{particle}} = \frac{0.101}{\sqrt{v}} \cdot 10^{-10} \text{ m}$$

$$\lambda_e > \lambda_p > \lambda_{\text{particle}}$$

HUP:

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

Heisenberg:

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} + \frac{8\pi^2 m}{h^2} [E - V] \psi = 0$$

Quantum Numbers:

$n \rightarrow$ Size

$l \rightarrow$ Shape

$m \rightarrow$ Orientation

$s \rightarrow$ Spin

Orbital Angular Momentum: $OAM = \sqrt{l(l+1)} \left(\frac{h}{2\pi} \right) = \sqrt{l(l+1)} \hbar$

Spin Angular Momentum: $SAM = \sqrt{s(s+1)} \left(\frac{h}{2\pi} \right)$

(Spin Magnetic Moment) $S_{\text{Magnet}} = \sqrt{n(n+2)}$

Max No. of electrons in a Shell $= 2n^2$

Max No. of electrons in a Subshell $= 2[2l+1]$

No. of Orbitals in Subshell $= [2(l+1)]$

No. of Subshells in Shell $= l$

Max No. of Orbitals in Shell $= n^2$

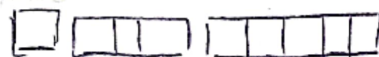
Nodes:

i. Radial Node

ii. Angular Node

Shells $\Rightarrow K, L, M, N, O, P, Q$

Subshells $\Rightarrow s, p, d, f$



$$R.N = n - l - 1$$

$$A.N = l$$

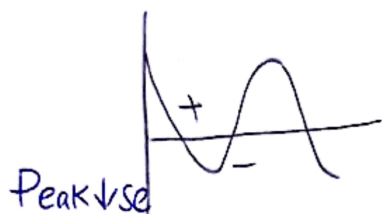
$$T.N = n - 1$$

orbitals

ψ : Wave function.

ψ^2 : Max Probability of finding an Electron.

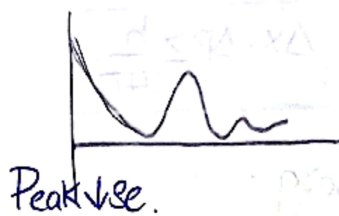
$\psi \propto \delta$



Cosine $\rightarrow$ S

Sine $\rightarrow$ P or d

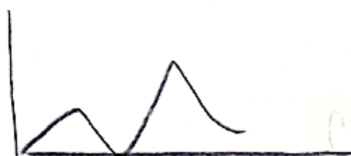
$\psi \propto r$



Cosine $\rightarrow$ S

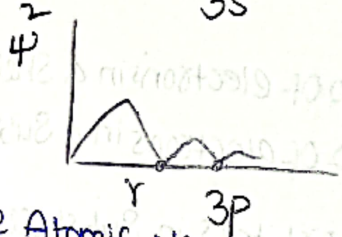
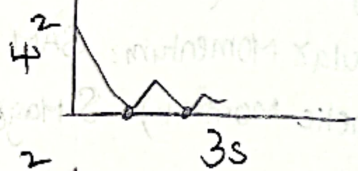
Sine $\rightarrow$ p or d.

$$4\pi r^2 dr \psi^2$$



Sine

Peak ↑ se.



Diff Elements having Same NO. of neutrons.

Isotopes

Isobars

Isotones

Isodiaphers

Isoelectronic

Isosters

: Same Atomic NO. but different Mass Numb
: Same Mass NO. but different Atomic Numb
: Same NO. of Neutrons.

: Same Isotopic numbers [Isotopic NO. = Mass NO. - 2 x A.F.N.]

: Same NO. of electrons

: Same NO. of Atoms and Same NO. of Electrons.

Mass of Neutron = 1.675×10^{-27} kg

Mass of Electron = 9.108×10^{-31} kg.